

Bicomplex Signal Geometry

Proof Companion and Coverage

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companion 0.01; local working checkpoint

This companion accompanies the v13 working revision. It records what can be replayed and where the mathematical proof remains outside Lean. The historical authority is the unchanged 43-page final v12 PDF, SHA-256

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Its physical and printed page numbers agree; Section 7 is on pages 14–16. The claim map includes all 66 named blocks: 47 definitions/results and 19 remarks. Original statements, actual domains, repairs, proof locations and remaining obligations are recorded separately.

1 The mathematical correction

The completed source has five rank-loss labels, including two endpoint corners omitted in v12. Actual derivative determinants establish the applicable smooth cross-cap recognition hypotheses. The polar observation map has genuine fold coordinates, independently of its incomplete Taylor remainder. Actual small-circle winding and exponential-cover lifting establish square-root sign change. The real link, weighted orbit and actual curvature exceptional cusp are distinct constructions.

The revised Mellin result fixes the measure and inner-product convention and includes the endpoint term. On $L^2(dx)$ its formal adjoint parameter is $1 - \sigma$; on $L^2(x^{2\beta-1}dx)$ it is $2\beta - \sigma$. The logarithmic unitary gives the actual half-line closure and adjoint domains H_0^1 and H^1 , with deficiency indices $(1, 0)$. The Gram entries are actual integrals. Every finite distinct frequency family is independent for every $T > 0$; separation bounds give conditioning and log-determinant estimates.

Capacity is proved by logarithmic cutoffs at all five labels. A different, weighted radial Hardy estimate controls L^2 mass near each singularity by $O(\varepsilon^2)$ times the form norm. Ordinary Rellich compactness on the regular complement then proves actual compact form embedding and a positive fixed-Dirichlet gap, including both endpoint corners.

The old compact-core minimum leaves infinitely many outer-boundary deficiency and harmonic modes. A collar extension followed by the fixed-Dirichlet resolvent injects arbitrary smooth regular-boundary data into those spaces. Thus the old $(2, 2)$ and exhaustive $U(2)$ classification fail as defined. The actual closed positive minimum still has a Krein extension and discrete reduced positive spectrum; its kernel is infinite-dimensional. The applicable abstract theorem gives the weak buckling correspondence on its actual domain. A finite point restriction is a different, conditional construction requiring graph-continuous surjective point traces.

2 Formal proof sources and exact replay

The Lean project uses Lean 4.33.1 and pinned Mathlib commit

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It reuses unchanged RuledV5 and StokesV5 proof sources after matching the map, domain and orientation. Its 27 public named theorems are audited as an own inventory; the 224 Ruled and 258 Stokes inventories are checked separately. The verifier rejects holes, custom axioms, unsafe declarations and `native_decide`; it permits only `propext`, `Classical.choice` and `Quot.sound` in the kernel audit.

Source	Coverage and its boundary
<code>Geometry.lean</code>	Actual completed-source rank classification, endpoint derivative, genuine polar fold, actual dipole winding and corrected parity. Standard cross-cap fibers, lift injectivity, residual and metric identities are also encoded. Smooth cross-cap recognition, link topology and capacity integrals remain written/external.
<code>Monodromy.lean</code>	Continuous actual square-root lift through the exponential covering, lifted endpoint and sign change for winding ± 1 . It applies directly to normalized actual small observation circles. Uniqueness of every continuous branch and double traversal remain written.
<code>Spectral.lean</code>	Actual complex Mellin interval integral with boundary term, zero-endpoint pairing, actual Gram entry integrals, modulus bound and finite-matrix eigenvalue enclosure. Functions have continuous given derivatives; $T > 0$ and separated finite frequencies are retained. Hilbert domains, Vandermonde independence, Hermitian operator-norm translation and logdet proofs remain written.
<code>BoundaryAlgebra.lean</code>	Selected scalar identities for transmission, real transgression, parity obstruction, cusp factorization and eta pairing. These do not supply an unbounded Dirac trace operator, sheaf category, Green parametrix or analytic eta continuation.

The independent SymPy replay rebuilds actual S and endpoint derivatives, cross-cap determinants, the polar remainder and fold identity, actual normal- plane jet invariants and cusp derivative criterion. It also checks the weighted formal-adjoint coefficient, standard $\Delta \log \rho$, a nonzero outer-boundary witness, the weight-zero cutoff commutator and a metric-shear counterexample. The unchanged three historical Bicomplex verifiers remain separate. Exact computation certifies these expressions, not the infinite- dimensional consequences.

3 The remaining analytic construction

The actual fixed-Dirichlet point restriction needs bounded graph traces $\tau : \text{Dom } H_F \rightarrow \mathbb{C}^2$ and a dense kernel. A two-sided Green parametrix must prove continuous resolvent correction for the displayed local L^p remainder, identify exhaustive maximal-domain coefficients and control overlap errors in specified graph spaces.

For the Dirac problem, specify the spin/Pin convention, singular-cut domains, flat phase-line fiber identification and a Clifford-compatible unitary cut map. A spin Pin lift alone supplies no map of those separate phase fibers. The regular-bank scalar Green calculation is conditional on

these data. Under dilation the normal cutoff commutator has weight zero; shrinking the patch alone gives no small Neumann-series remainder.

The chosen complex A_3 completion has explicit compactification and divisor, so external Milnor, Gysin and mixed-Hodge theorems have stated input objects. Normalization support/recollement and Clifford/Pin results remain external. These results, compactness and operator-domain proofs are not formalized by the present inventory. The current checkpoint is partial coverage; its manifest or successful replay does not make it full-paper Lean coverage.

4 Replay and artifact provenance

From the repository root, use the registered interpreter, without Python `-O`:

```
local/cache/python/legacy-reconstruction-venv/bin/python -B \
verification/legacy-reconstruction/verify_all.py
```

Then run the new exact replay and formal verifier:

```
bicomplex_dir=papers/legacy-geometry/bicomplex-signal-manifolds
python -B "$bicomplex_dir/verification/verify_revision.py"
python -B companions/lean/bicomplex-signal-manifolds/verify_lean.py
```

Use the same environment with SymPy for the exact script, and the pinned Lean toolchain for the formal verifier. Run each dependency's own `verify_lean.py` separately. The claim-map builder re-extracts the hash-locked PDF and rejects missing or duplicate named entries.

The standalone manuscript and this source are checked by the native LaTeX editor. Exported PDFs receive separate pagewise visual QA and SHA-256 records. The checkpoint receipt distinguishes native source compilation, exported PDF comparison, exact calculation and kernel audit. Original-location and ZIP-extracted replay take place on the same host with the registered tools; they are not an independent-host certificate. Historical PDFs, ancestor figures, sealed packages and their receipts are preserved.

A Public named theorem inventory

All names below have the namespace `Bicomplex`. Precise types are in the source and the Markdown coverage table.

```
actual_source_rank_classification; actual_endpoint_rank; actual_polar_fold;
actual_dipole; observation_field_even; normalized_polynomial_derivative;
crosscap_lift_injective; crosscap_actual_fibers; crosscap_metric_determinant;
crosscap_residual_positive; square_root_lift; log_lift_endpoint;
unit_winding_sign_change; local_observation_root_sign; mellin_interval_green;
mellin_zero_boundary_pairing; formal_coefficient_symmetry; gram_diagonal;
gram_entry_integral; gram_entry_bound; gram_eigenvalue_enclosure;
zero_boundary_core_witness; transmission_unit_modulus_iff;
real_transgression_coefficient; sign_intertwiner_zero; cusp_return_square;
half_integer_pairing.
```